UNIT – I : SOIL EXPLORATION

Foundation Engineering (23A01604c) · Handwritten Notes

1.1 Introduction

- Every structure ends up on the ground, so the **soil must carry the load safely.**
- Soil exploration (= site investigation / sub-surface exploration) is the process of finding the **soil profile, soil properties and ground water table** at a site.
- It is the **first step** of foundation design – cheapest part of the job (0.5 – 1 % of project cost) but a wrong investigation can cost many times more.

> **Definition:** Soil exploration is the planned collection of information on the sub-soil – its stratification, engineering properties, GWT and rock level – by field and laboratory methods so that a safe and economical foundation can be designed.

1.2 Need for Soil Exploration

1. To know the **nature of soil deposits** – sequence, thickness, extent of the strata.
2. To find the **depth of ground water table** (GWT) and its fluctuation.
3. To determine **engineering properties** – shear strength (c, φ), compressibility (C_c, m_v), permeability, density.
4. To select the **type and depth of foundation** (shallow / pile / well / raft).
5. To estimate **safe bearing capacity** and probable **settlement.**
6. To predict & prevent problems – swelling, collapse, liquefaction, scour, aggressive soil/water (sulphates, chlorides).
7. To select **construction method** – dewatering, shoring, bracing of excavations.
8. To investigate **failure / distress** of structures, slopes, dams; to locate borrow areas.

1.3 Stages of Investigation

- **Reconnaissance** – desk study, geological maps, site walk-over, existing records.
- **Preliminary exploration** – few borings, to find general soil profile, rock depth, GWT.
- **Detailed exploration** – close spaced borings, sampling and in-situ tests.

1.4 Planning of the Exploration Programme

The programme depends on the type of structure, its importance, loads and the soil variability. Steps:

1. Collect **data on the proposed structure** – type, size, column loads, tolerable settlement.
2. Collect **data of the site** – topography, geology, nearby structures, past land use (fill, mine, tank).
3. Fix **number, location, depth** of boreholes / pits.
4. Choose **method of boring and sampling**; decide in-situ tests (SPT, CPT, vane, PLT).
5. Decide **laboratory tests** to be done on samples.
6. Collect results → prepare **borelogs, soil profile, report.**

(a) Depth of exploration

- Boring must go to the depth where the **stress increase $\Delta\sigma$ is only ≈ 10 % of the foundation pressure q** (i.e. the "significant depth"), or until hard rock/stratum.
- IS 1892 : at least **1.5 × width** of foundation (isolated); deeper for strip/raft.

Foundation / structure	Minimum depth of boring
Isolated square footing (width B)	≈ 1.5 B – 2 B below foundation level
Strip footing (long)	≈ 3 B – 4 B
Raft (width B)	≈ 1.5 B (and ≥ 6 m) below raft
Pile foundation	≥ 5 m below pile tip / 1.5 × group width
Highway cutting / embankment	≈ 1 – 2 × height; ≥ 3 m below subgrade

(b) Spacing of boreholes

Project	Spacing (m)
Multi-storey buildings	10 – 30
One-storey industrial plants	20 – 60
Highways	150 – 300
Earth dams	15 – 60

- Minimum **3 boreholes** not in a straight line (to get dip of strata); more when soil is erratic.

(c) Typical layout of boreholes

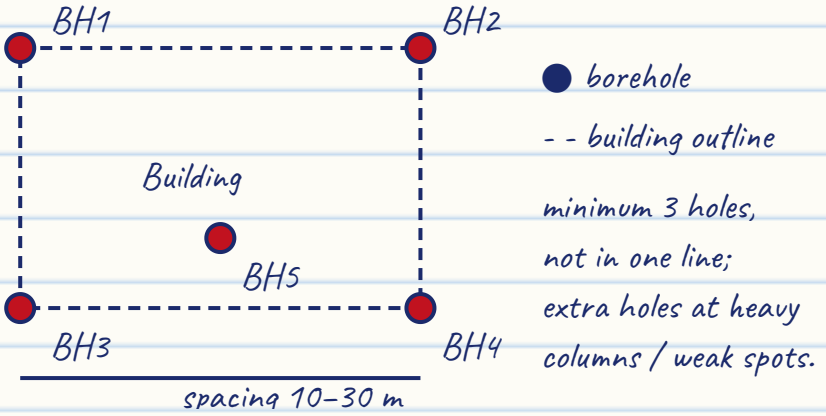


Fig 1.1 Borehole layout for a building

I.5 Methods of Soil Exploration

Classification:

- **Direct methods** – open excavations: test pits, trenches, shafts, drifts.
- **Semi-direct methods** – borings (auger, wash, percussion, rotary) with sampling.
- **Indirect methods** – geophysical (seismic, electrical resistivity), sounding/penetration (SPT, CPT, DCPT), vane.

A. Open excavations (direct)

- **Test pits** – 1.2 m × 1.2 m (min) or larger; depth up to 3 – 4 m (above GWT).
- **Trenches** – long narrow cuts for strata continuity (e.g. along a pipeline, fault).
- **Shafts / drifts** – large; used for dam sites in rock.

Advantages:

- Soil is visible **in-situ** → exact stratification, fissures, lenses seen; best undisturbed (block) samples; in-situ tests like PLT possible.

Limitations:

- Costly and slow beyond 3 – 5 m; needs shoring/dewatering below GWT; unsafe in loose sand/soft clay.

B. Boring methods

Used when the depth is large. Hole diameter 100 – 200 mm (IS 1892).

I. Auger boring

- **Hand auger** (post-hole / helical) – 100 – 200 mm dia; depth ≤ 5 – 6 m; for soft to stiff cohesive soils above GWT. Soil is brought up in the auger – disturbed sample.
- **Power (flight) auger** – continuous flights rotated by machine; depth up to 30+ m in cohesive soil; hollow-stem type allows SPT/sampling through the stem.

- Not suitable in gravel, boulders, below GWT in loose sand (hole collapses).

2. Wash boring

- Casing (hole lining) is driven; a **chopping bit** on hollow drill rod loosens soil while **water is jetted** through the rod.
- Water + cuttings rise in annulus to a settling tank; cuttings show the soil change only roughly.
- Suitable for sand, silt and soft clay; not for gravel / boulders / hard clay. Cheap, quick; SPT can be done at intervals.

3. Percussion (cable-tool / churn) drilling

- A heavy chisel bit is repeatedly **lifted and dropped** on a cable; soil/rock is broken, mixed with water to slurry and removed by a bailer/sand pump.
- Good for hard clay, gravel, boulders, soft rock; casing advanced simultaneously. Slow. Samples are crushed → disturbed.

4. Rotary drilling

- A **rotating bit** (drag/ roller/ diamond) on drill rods cuts the soil or rock; drilling fluid (bentonite mud / water) is pumped down, carries cuttings up and keeps the hole open.
- Fast, can go very deep; **core barrel** with diamond bit gives **rock cores** (NX, BX, AX sizes).
- Used for rock, hard soil, deep investigations (dam, bridge sites); cost is high.

Comparison of boring methods

Method	Soil type	Depth	Sample
Hand auger	soft/stiff clay, above GWT	≤ 6 m	disturbed
Power auger	clays, silts, dense sand	≤ 30 m	disturbed
Wash boring	sand, silt, soft clay	≤ 30 – 60 m	cuttings
Percussion	all soils incl. gravel, soft rock	deep	disturbed
Rotary	all soils & rock	very deep	cores

C. Geophysical (indirect) methods

Measure physical properties (velocity, resistivity) from the surface – **quick, cheap, no sample**; results must be calibrated with a few borings.

(i) Seismic refraction method

- Shock wave created by hammer/explosive at surface; **geophones** record first arrival times at different distances.
- Time–distance plot gives wave velocities V_1, V_2 of layers and the thickness of top layer:

$$H = (x_c / 2) \cdot \sqrt{[(V_2 - V_1)/(V_2 + V_1)]}$$

x_c = critical distance (break in the T–x graph). Needs $V_2 > V_1$.

(ii) Electrical resistivity method (Wenner)

- Four equally spaced electrodes (spacing a) – current I through outer two, potential V across inner two.

$$\rho = 2\pi a V / I$$

- Resistivity changes with moisture, salt content & soil type: **sharp drop at GWT**; high for rock/dry gravel, low for clay. Used for location of GWT, rock, buried channels, corrosion studies.

1.6 Soil Sampling

Purpose: to get soil specimens for visual classification and lab tests (grain size, Atterberg limits, c, φ, consolidation, permeability).

Classification of samples

- **Disturbed – non-representative:** soil mixed with other strata / washed fines (e.g. wash cuttings). Only for rough identification.
- **Disturbed – representative:** structure destroyed but all constituents and water content retained (split-spoon, auger samples, kept in air-tight jar). Used for grain size, Atterberg limits, specific gravity, compaction.
- **Undisturbed:** structure, density, water content unchanged as far as possible (thin-walled tube, block samples). Needed for **shear strength, consolidation, permeability** tests on clays.

No sample is truly "undisturbed" – sampling disturbance is only minimised.

Sampling methods

1. **Auger / test-pit samples** – disturbed.

2. **Chunk (block) samples** – cut by hand from pit, coated with paraffin wax, wrapped & boxed – best quality, undisturbed.
3. **Split-spoon (SPT) sampler** – disturbed representative; 35 mm ID.
4. **Open drive (thin-wall Shelby tube)** sampler – undisturbed in soft–medium clays; tube pushed at a steady rate.
5. **Stationary piston sampler** (Osterberg / Hvorslev) – piston keeps soft clay from entering before the tube is pushed; better recovery in soft clays.
6. **Rotary (Denison, Pitcher) samplers** – double-tube with a rotating outer cutting shoe, for stiff / hard clay and soft rock.
7. **Sand sampling** – hard to sample; core-catcher / freezing technique, or in-situ density by sand-replacement & core-cutter methods.

Handling of samples

- Seal both ends with wax, label (project, BH no., depth, date), keep upright, protect from sun, vibration and drying; test early.

1.7 Design Features of a Sampler

Sample disturbance depends on the **geometry of the cutting edge**. Let

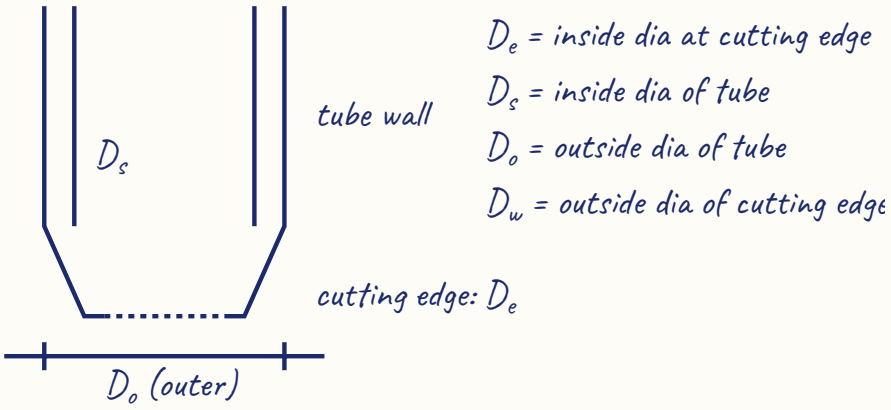


Fig 1.3 Thin-walled tube sampler – shoe details

$$\text{Area ratio } A_r = (D_w^2 - D_e^2)/D_e^2 \times 100\ \%$$
$$\text{Inside clearance } C_i = (D_s - D_e)/D_e \times 100\ \%$$
$$\text{Outside clearance } C_o = (D_w - D_o)/D_o \times 100\ \%$$
$$\text{Recovery ratio } L_r = L / H \ (L = \text{sample length}, H = \text{penetration})$$

- A_r = volume of soil displaced ÷ volume of sample. For undisturbed sample $A_r \leq 10\ \%$ (thin-wall; IS 2132 $\leq 20\ \%$ for thick-wall tubes).

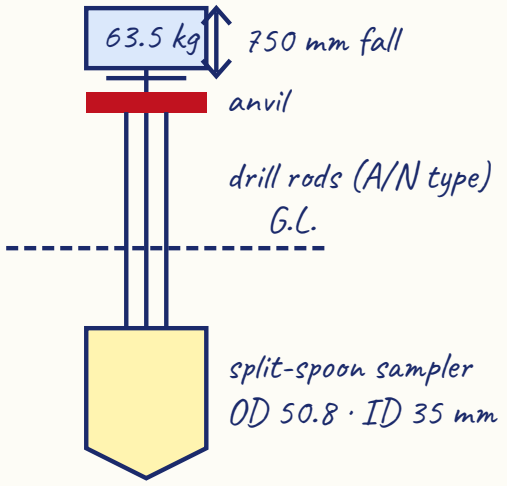
- C_i = 0.5 – 3 % : lets sample expand slightly, reduces wall friction (small clearance for clays, up to 3 % for long tubes).
- C_o = 0 – 2 % : eases withdrawal of the tube.
- **Recovery ratio** should be ≈ 96 – 98 % (≥ 100 % means swelling, < 100 % shortening).
- Cutting edge angle small (≈ 10°–12° for thin tubes), tube length ≈ 10 – 20 × D, a **vent / ball valve** at the head to release air and hold the sample (suction).

Split-spoon sampler: A_r ≈ 110 % (50.8 & 35 mm) – hence **disturbed**.

Field Tests

1.8 Standard Penetration Test (SPT) – IS 2131

Most widely used in-situ test, especially in **cohesionless soils** where undisturbed sampling is difficult.



① seating drive 150 mm
② test drive 2 × 150 mm
→ blows counted = N
③ stop if N > 50 (refusal)

Fig 1.4 SPT assembly

Procedure

1. Bore hole cleaned to required depth (150 – 200 mm dia); casing used if sides cave.
2. Split-spoon (length 450 mm) lowered at the bottom with drill rods.
3. Hammer of **63.5 kg (140 lb)** dropped freely through **750 mm** on the anvil.
4. Sampler driven **450 mm** in 3 steps of 150 mm: blows for the **first 150 mm are neglected** (seating).
5. N-value = blows for the last 300 mm (2nd + 3rd drive). If 50 blows do not give 150 mm → **refusal**, record penetration.
6. Sampler is withdrawn, split opened and soil is taken as a representative sample.

7. Test done every **1.0 – 1.5 m** depth or at change of stratum.

N is the number of blows per 300 mm penetration (blows/ft in the old units).

SPT corrections

(1) Dilatancy (water-table) correction – for fine / silty sand below GWT (Terzaghi & Peck), when observed $N > 15$:

$$N_c = 15 + \tfrac{1}{2}(N - 15)$$

(excess pore pressure in dense fine sand makes N too high; no correction if $N \leq 15$ or in coarse sand/gravel).

(2) Overburden pressure correction – N increases with depth, so reduce to a standard σ'_o:

$$N' = C_N \cdot N \text{ with } C_N = 0.77 \log_{10}(2000 / \sigma'_o) \; [\sigma'_o \text{ in kN/m}^2,\ \geq 25]$$

$$\text{(Liao–Whitman: } C_N = \sqrt{(100/\sigma'_o)}\text{)}$$

(3) Energy / other corrections (modern practice, N_{60}):

$$N_{60} = N \cdot \eta_H \eta_B \eta_S \eta_R / 0.60$$

- η_H hammer efficiency (safety hammer 0.6–0.75, donut 0.45–0.6, automatic 0.8–1.0)
- η_B borehole dia (1.0 for 65–115 mm, 1.05 for 150 mm, 1.15 for 200 mm)
- η_S sampler (1.0 with liner, 1.2 without liner)
- η_R rod length (0.75 for < 4 m, 0.85 for 4–6 m, 0.95 for 6–10 m, 1.0 for > 10 m)

Order of applying (IS): first dilatancy correction, then overburden correction (apply dilatancy only when $N > 15$ in saturated fine sand).

N-value correlations (Terzaghi & Peck)

Sand N	Compactness	Dr %	φ
0–4	very loose	<15	<28°
4–10	loose	15–35	28–30°
10–30	medium	35–65	30–36°
30–50	dense	65–85	36–41°
>50	very dense	>85	>41°

Clay (N → consistency, q_u kN/m²): <2 very soft (<25); 2–4 soft (25–50); 4–8 medium (50–100); 8–15 stiff (100–200); 15–30 very stiff (200–400); >30 hard (>400).

Worked Example 1 – SPT corrections

SPT in fine saturated sand at 6 m depth gave $N = 32$. $\gamma_{sat} = 19$ kN/m³, GWT at 2 m below GL, $\gamma_w = 10$ kN/m³. Find the corrected N.

- $\sigma'_o = 2(19) + 4(19 - 10) = 38 + 36 =$ **74 kN/m²** (taking γ above GWT also 19)
- Dilatancy: $N_c = 15 + 0.5(32 - 15) = 15 + 8.5 =$ **23.5**
- $C_N = 0.77 \log(2000/74) = 0.77 \times 1.432 =$ **1.10**
- $N' = 1.10 \times 23.5 =$ **25.9 ≈ 26** → medium dense sand ($\varphi \approx 33°$)

Uses of SPT results: relative density, φ, bearing capacity (Meyerhof, Peck), settlement of footings, pile capacity, liquefaction check, c_u of clays ($c_u \approx 6\,N$ kN/m²).

Merits / limitations of SPT

- \+ Simple, cheap, gives sample, done in any borehole, wide database.
- – Poor repeatability (hammer type, operator, rod energy); disturbed sample; not reliable in clays, gravels (cobble gives false high N).

1.9 Dynamic Cone Penetration Test (DCPT) – IS 4968 (Part 1 & 2)

- A **cone (60° apex)** is driven by a **65 kg hammer, 750 mm fall**; no sample; used mainly in sands/gravels.
- **Type A (50 mm cone):** blows per **100 mm** penetration = N_{cd}. Used up to ~ 8 m with no casing.
- **Type B (65 mm cone) + bentonite slurry:** blows per **300 mm** = N_{cd}; slurry reduces rod friction, deep up to 20 m+.
- Continuous record of penetration resistance with depth; correlated with SPT: $N_{cd} \approx N$ (approx.) – found for a site by calibration.

1.10 Static Cone Penetration Test (SCPT / CPT) – IS 4968 (Part 3)

Also called **Dutch cone test.** A cone is pushed (not driven) into the ground at a constant rate of **20 mm/s** and the resistance is measured.

- **Cone resistance** $q_c = Q_c / A_c$ ($A_c = 10$ cm²)
- **Sleeve (skin) friction** $f_s = Q_s / A_s$ ($A_s = 150$ cm²)
- **Friction ratio** $R_f = f_s / q_c \times 100\,\%$ → identifies soil: sand $R_f < 1\,\%$, clay 3 – 5 %.

- Electric cone / **piezocone (CPTu)** measures q_c, f_s and pore pressure u continuously.

Applications / correlations

- Sands: $q_c \approx 400\ N$ (kN/m²) (N = SPT); φ from q_c charts; bearing capacity and settlement (Schmertmann).
- Clays: $c_u = (q_c - \sigma_{vo}) / N_k$, $N_k = 15 - 20$.
- Pile capacity: end bearing ≈ q_c, skin friction ≈ f_s (Dutch method).

SPT vs CPT: SPT is dynamic, gives a disturbed sample, discrete readings; CPT is static, no sample, continuous record, not for gravel.

I.11 Vane Shear Test – IS 4434

An in-situ test to get the **undrained shear strength c_u of soft, saturated clays** and their **sensitivity** (no sampling disturbance).

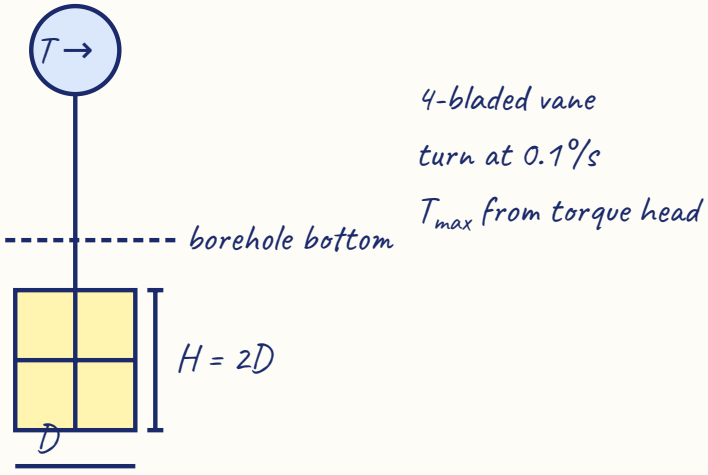


Fig 1.7 Vane shear apparatus

- Vane (two orthogonal blades, typ. D = 75 mm, H = 150 mm, so H/D = 2) is pushed to the test depth, then **torque** applied at ≈ 0.1°/s till failure of the cylinder of soil. Peak T → c_u.
- Rotate rapidly 10 times – **remoulded** strength measured again.

Torque resisted by cylindrical surface + top & bottom circular ends:

$$T = \pi D^2 H c_u/2 + 2 \cdot (\pi D^3 c_u/12) \Rightarrow c_u = T / [\pi (D^2H/2 + D^3/6)]$$

$$H = 2D : c_u = T / (3.66\ D^3)$$

$$\text{Sensitivity } S_t = c_u(\text{undisturbed}) / c_u(\text{remoulded})$$

- Suitable for soft clay (c_u < 100 kN/m²); not in sandy/ stony or stiff clays; correct c_u for plasticity (Bjerrum μ) for embankment stability.

Example: $D = 75$ mm, $H = 150$ mm, $T = 38$ N·m: $c_u = 38 / (3.66 \times 0.075^3) = 38 / (1.544 \times 10^{-3}) =$ **24.6 kN/m²** (soft clay).

1.12 Plate Load Test (PLT) – IS 1888

A field loading test on a small rigid plate placed at foundation level to obtain the **load–settlement behaviour**, ultimate bearing capacity and the modulus of subgrade reaction.

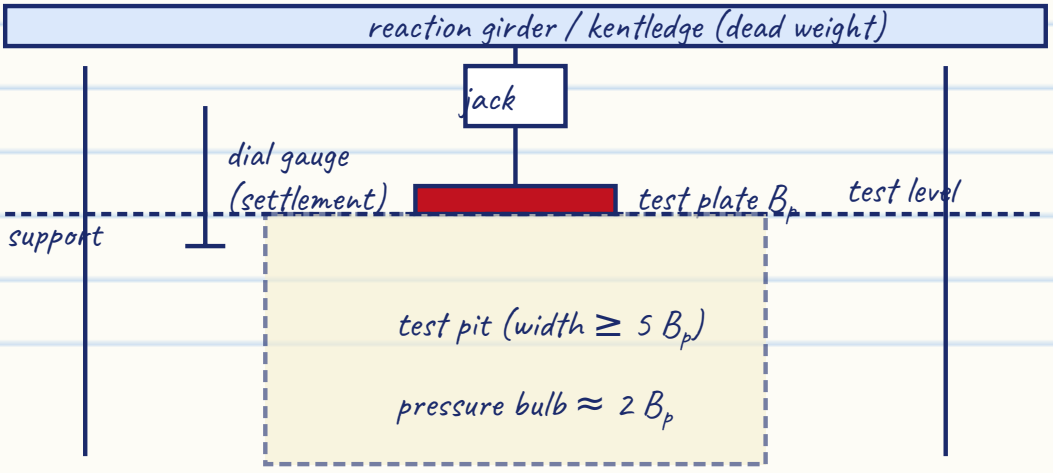


Fig 1.8 Plate load test set-up (gravity loading)

Equipment & procedure

- Plates: **mild steel, 25 mm thick, 300 – 750 mm square or circular** (300, 450, 600, 750 mm); sand bedding.
- Test pit width ≥ 5 B_p; bottom levelled at proposed foundation level; GWT lowered if above.
- Load applied by **hydraulic jack** against kentledge (or anchor / truck reaction); settlements by **2 – 4 dial gauges** (0.01 mm) on independent datum beam.
- Seating load ≈ 7 kN/m² released; then **increments = 1/5 of estimated safe load** (≈ 1/10 of ultimate).
- Each increment kept until **rate of settlement ≤ 0.02 mm/min** (or 1 h in clays); readings at 1, 2.25, 4, 6.25, 9, 16, 25… min.
- Continue to **failure** or to **1.5 – 2 × the design load** / settlement of 25 mm.

Load–settlement curve & ultimate load

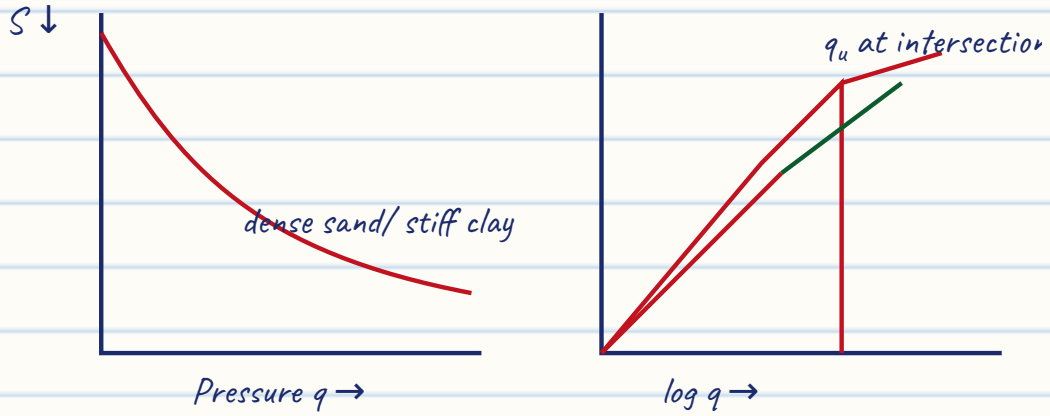


Fig 1.9 Load–settlement curves; ultimate load from log–log plot (two straight lines intersect)

- **Clear failure** (dense sand, stiff clay): peak/ break of curve = q_u.
- **No clear failure** (loose sand, soft clay): double tangent or log q – log S intersection; or load at S = 25 mm (or 10 % of B).

Size (scale) effect – Terzaghi & Peck

Footing size B_f and plate B_p:

Soil	Bearing capacity	Settlement
Clay (cohesive)	$q_{uf} = q_{up}$ (independent of size)	$S_f = S_p \cdot (B_f / B_p)$
Sand (cohesionless)	$q_{uf} = q_{up} \cdot (B_f / B_p)$	$S_f = S_p \cdot [B_f (B_p + 0.3) / (B_p (B_f + 0.3))]^2$ (B in m)

For c–φ soils, the sand rule is used on the safe side. Safe pressure = q_u / FS (2.5 – 3) with S ≤ allowable (25 mm isolated footing in sand).

Worked Example 2

PLT on sand with 0.3 m square plate: q_{up} = 180 kN/m², settlement 10 mm at design pressure. Footing 1.5 m square. Find q_{uf} and S_f.

- q_{uf} = 180 × 1.5/0.3 = **900 kN/m²**; with FS = 3 → q_{safe} = **300 kN/m²**
- S_f = 10 × [1.5 (0.3 + 0.3) / (0.3 (1.5 + 0.3))]² = 10 × (0.9/0.54)² = 10 × 2.78 = **27.8 mm** → if > 25 mm, reduce pressure

Limitations of the plate load test

- Influences only ≈ 2 B_p depth – cannot sense deeper weak layers (clay below sand) that the large footing stresses.
- Does **not** give long-term consolidation settlement of clays (short duration test).

- Size effect – results must be corrected; GWT effects; costly and time consuming.
- Test the **sub-grade modulus:** $k_s = q / S$ (kN/m³) for plate of 0.3 m; footing: $k_f = k_p (B_p/B_f)$ for clay; $k_f = k_p [(B_f+0.3)/(2B_f)]^2$ for sand.

1.13 Pressuremeter Test (Ménard)

A cylindrical probe is expanded **radially** inside a borehole; the pressure–volume change gives **in-situ stress-strain modulus and strength** of soil/soft rock.

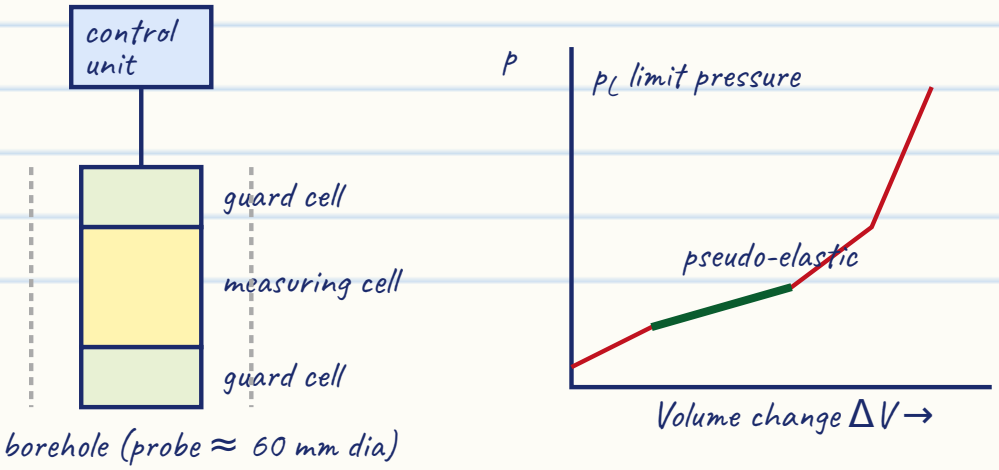


Fig 1.10 Ménard pressuremeter & pressure–volume curve

- Probe: 3 cells – central **measuring cell** (water-filled) + two **gas-filled guard cells** (keep the strain plane-cylindrical).
- Pressure raised in equal steps (each 1 min), volume change read at 15, 30, 60 s.

Results of pressuremeter

$$E_m = 2(1+\mu)(V_0 + V_m) \cdot \Delta p / \Delta V$$

- V_0 = initial volume of measuring cell, V_m = volume at middle of the straight (pseudo-elastic) portion, μ = Poisson's ratio (≈ 0.33), $\Delta p/\Delta V$ = slope of that portion.
- p_0 – at-rest horizontal pressure (start of straight part); p_f – creep pressure (end of straight part); p_L – limit pressure (volume = 2 V_0 … cavity doubles).
- Bearing capacity (Ménard): $q_u = \sigma_{v0} + k(p_L - p_0)$, k = bearing factor (0.8 – 3) depending on soil & D_f/B; safe $q = q_u/3$.

Advantages: in-situ, gives modulus + strength, works in all soils & weak rock. Limitations: borehole disturbance, costly, interpretation empirical.

1.14 Soil Investigation Report

Final document handed to the designer (IS 1892): must be **clear, complete, honest**, with data separated from opinion.

1. **Introduction** – project, client, purpose, dates, scope of work.
2. **Site description** – location, topography, geology, nearby structures, climate, drainage.
3. **Field investigation** – number, location, depth of boreholes/pits, method, in-situ tests, sampling, GWT observed.
4. **Laboratory tests** – classification, c, φ, C_c, permeability, chemical tests, rock tests.
5. **Soil profile** – borelogs, cross-sections, strata description.
6. **Analysis & recommendations** – type and depth of foundation, **safe bearing capacity**, settlement, ground improvement, dewatering, precautions.
7. **Annexures** – site plan with BH locations, borelogs, test data sheets, graphs, photos.

Important Questions (Unit I)

1. What is the need for soil exploration? Explain the stages and planning of an exploration programme. (5–10 M)
2. State the depth & spacing of boreholes for (a) building, (b) highway, (c) dam.
3. Describe the different methods of boring with neat sketches – auger, wash, percussion, rotary.
4. Explain the **geophysical methods** – seismic refraction and electrical resistivity. Mention merits and demerits.
5. Differentiate disturbed and undisturbed samples. How are undisturbed samples obtained?
6. Define **area ratio, inside clearance, outside clearance**. What values are recommended for undisturbed sampling?
7. Describe the SPT with sketch. List the corrections applied to N and their formulas. (very frequent)
8. Write short notes on DCPT (type A and B) and static cone test; compare with SPT.
9. Explain the vane shear test and derive $c_u = T / [\pi (D^2H/2 + D^3/6)]$.
10. Describe PLT procedure and how load-settlement curve is used; give size-effect formulas. State limitations.
11. Explain the pressuremeter test and the parameters obtained.
12. List the contents of a soil investigation report. Draw a typical borelog.

Numerical practice

- SPT N = 40 at 8 m in fine saturated sand: find N_c, N' (γ = 20, GWT at GL, take γ_w = 10).

- Sampler: $D_o = 54$, $D_i (D_s) = 50$, $D_e = 49$ mm, $D_w = 53$ mm … find A_r, C_r, C_c.
- Vane $D = 50$ mm, $H = 100$ mm, $T = 14$ N·m → c_u and sensitivity (remoulded $T = 3.5$ N·m).
- PLT 0.45 m plate: 25 mm settlement at 200 kN/m² in sand; find settlement of 3 m footing at same pressure.

★ Quick revision: SPT = 63.5 kg × 750 mm, N = last 300 mm · Cone = 60°, 10 cm², 20 mm/s · Vane $c_u = T/3.66D^3$ · PLT depth of influence = $2B_p$ · $A_r \leq 10\ \%$